

Automated Test Report

StockSense Blazor application and Gemini chatbot microservice - fresh local verification executed on 30 July 2026.

PASS WITH LIMITATIONS

Executive summary

All executable chatbot unit tests and all enabled .NET tests passed. The solution also compiled with zero errors. The status is qualified because 12 SQL Server integration tests were skipped, chatbot database and Gemini interactions were mocked, and no browser-to-API end-to-end run was included.

92/92

Chatbot tests passed

85/85

Enabled .NET tests passed

0

Build errors

12

Integration tests skipped

Release interpretation

The tested code paths are healthy under mocked dependencies and the available local test configuration. Before production sign-off, run the skipped SQL tests and a controlled live Gemini/database/API smoke test.

Tested components

Component	Verification	Result
Gemini chatbot	Python unittest discovery: 92 tests	PASS
Python source	py_compile for main.py and admin_tools.py	PASS
StockSense solution	dotnet build StockSense.slnx --no-restore	PASS - 9 warnings
StockSense tests	xUnit: 97 discovered, 85 passed, 12 skipped	PASS WITH SKIPS

Chatbot automated test results

Command: uv run --with-requirements requirements.txt python -m unittest discover -s tests -v

Metric	Observed result
Outcome	PASS - exit code 0
Tests	92 passed; 0 failed; 0 errors; 0 skipped
unittest execution	0.096 seconds
End-to-end command	2.357 seconds including uv startup
Syntax compilation	main.py and admin_tools.py passed py_compile
Runtime	Python 3.12.13; uv 0.10.12
Model configuration	gemini-3.1-flash-lite default; environment override retained

Coverage by test category

Category	Tests	Examples of validated behavior
Appointment scheduling	35	Natural service matching, ambiguity, typos, date bounds, overlap, lead time, deterministic replies
Service catalog	11	Active-only results, aliases, pricing, bounded choices, malformed rows
Formatting and instructions	9	Chat formatting, Markdown cleanup, customer ID/quantity redaction
Admin routing	9	Role exposure, forced typed tools, Admin-only user counts, SELECT precedence
Admin report tools	8	Parameterized SQL, field allowlists, date/filter validation, row bounds
Pre-built packages	8	Public projections, compatibility scope, routing and safe fallback
Customer product policy	6	Available-only listings, named checks, out-of-stock-list refusal
Tool-loop safeguards	6	Execution budget, repeated calls, tool-free synthesis, safe fallbacks

Chatbot dependency limitation

Gemini model calls and SQL Server calls were mocked. These tests verify routing, query construction, privacy shaping, and deterministic behavior, but not live model quality, quotas, network behavior, database schema drift, or production performance.

StockSense .NET verification

Runtime: .NET SDK 10.0.302

Build results

Metric	Observed result
Command	dotnet build StockSense.slnx --no-restore
Outcome	PASS
Errors	0
Warnings	9
Duration	9.84 seconds

xUnit results

Metric	Observed result
Command	dotnet test tests\StockSense.Tests\StockSense.Tests.csproj --no-restore
Outcome	PASS
Discovered	97
Passed	85
Failed	0
Skipped	12
Duration	2.4876 seconds

Skipped SQL Server integration tests

Eight OrderSlipWorkflowSqlServerTests and four ProductInventoryUpdateSqlServerTests were skipped because STOCKSENSE_TEST_SQL_CONNECTION was not configured. These tests cover transactional receiving, duplicate/open-order controls, concurrency, inventory audit behavior, and database mutation safeguards.

Required follow-up

Configure a disposable SQL Server test database through STOCKSENSE_TEST_SQL_CONNECTION, then rerun the full .NET suite before treating database workflows as production-verified.

Build warnings

Warnings did not fail the build, but they remain actionable quality and security items.

Priority	Warning	Location / impact
High	NU1902 (2 occurrences): MailKit 4.15.0 has known moderate-severity vulnerability GHSA-9j88-vvj5-vhgr	Infrastructure and Client package graphs; assess and upgrade to a remediated version
Medium	RZ10012: BbPortalHost treated as unexpected markup	StockSense.Client/Layout/MainLayout.razor:40; verify namespace/import and component rendering
Medium	CS8602 possible null dereference (3 occurrences)	EmployeeAppointments.razor:174, :180 and OrderSlipHelper.cs:62
Low	CS0168 unused exception variable	StockSense.Web/Program.cs:182
Low	CS0649 field never assigned	Account Manage Index.razor:59
Low	CS0414 assigned field never used	Employee POS.razor:452

Verification boundaries

Area	Status	Boundary
Live Gemini API	Not executed	No live Flash-Lite latency, quota, safety filter, or semantic-quality validation
Live chatbot SQL	Not executed	No live driver, authentication, schema compatibility, data drift, or query-performance validation
HTTP/API integration	Not executed	No FastAPI TestClient/real server or Blazor-to-chatbot end-to-end flow
Browser/UI	Not executed	No automated keyboard, rendering, or cross-browser validation
Coverage percentage	Not measured	No line or branch coverage instrumentation; no percentage is claimed

Recommendations and sign-off

Order	Recommended action	Exit criterion
1	Run the 12 skipped SQL integration tests against a disposable SQL Server database	97/97 .NET tests execute with zero failures
2	Add a protected live chatbot smoke suite covering Customer and Admin roles	Verified role propagation, live queries, and safe output against controlled test data
3	Add HTTP end-to-end tests from Blazor proxy to FastAPI	Authentication, timeout, malformed-response, and role boundaries pass automatically
4	Resolve or formally accept the MailKit vulnerability warning	No unresolved NU1902 warning, or documented risk acceptance
5	Correct BbPortalHost and nullability warnings	Build completes without RZ10012 or CS8602 warnings
6	Introduce coverage reporting for both test suites	Baseline line/branch measurements tracked in CI

Final assessment

PASS WITH LIMITATIONS. No failures were observed in the 177 tests that executed across the chatbot and .NET suites. The remaining risk is concentrated in skipped/live integration paths and the nine build warnings.

Evidence summary

Chatbot: 92 passed, 0 failed, 0 skipped. .NET: 85 passed, 0 failed, 12 skipped. Combined executed tests: 177 passed. Solution build: 0 errors, 9 warnings. All commands were run fresh for this report; isolated .NET build/test output directories were removed after verification.

Report provenance

Generated from local automated test command output and source-level test inventory in the StockSense-System and Gemini_Chatbot workspaces. No external production systems were modified.